

Kaibalya Sahoo

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PROFESSIONAL SUMMARY

Software Engineer with a strong foundation in data structures, algorithms, and object-oriented design, complemented by hands-on experience building scalable web applications using React.js, Node.js, Nest.js, and PostgreSQL. Proven ability to design and ship secure, production-grade systems — including role-based access control, payment processing, and authentication — across internship and personal-project settings. Comfortable working across the full stack, from problem-solving and system design to frontend UI, backend architecture, database design, caching, and cloud deployment (AWS, Docker). Quick learner focused on writing clean, efficient, well-tested code and collaborating effectively in agile, cross-functional teams.

SKILLS

Core CS Fundamentals: Data Structures & Algorithms, Object-Oriented Programming, Time & Space Complexity Analysis, Problem Solving

Languages: JavaScript, TypeScript, Python

Frontend: React.js, Next.js, Tailwind CSS, Bootstrap, Responsive UI Design

Backend: Node.js, Express.js, Nest.js, FastAPI, RESTful APIs, WebSockets

Databases: PostgreSQL, MongoDB, QdrantDB (Vector Database), TypeORM

Cloud & DevOps: AWS (EC2, Lambda, ECS), Google Cloud Platform, Docker, Git, GitHub, Vercel, Netlify, Render, CI/CD

Architecture & Tools: Redis, BullMQ, Microservices, Authentication (JWT), Role-Based Access Control, Payment Integration (Razorpay), Webhooks

Generative AI: OpenAI, Gemini, LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Agentic AI

Testing & Practices: Vitest, Unit Testing, Agile/Scrum, Code Review, Technical Documentation

Soft Skills: Problem-Solving, Cross-Functional Collaboration, Communication, Adaptability, Ownership

PROFESSIONAL EXPERIENCE

Software Developer Intern — HyScaler *Mar 2026 – May 2026*

- Engineered a role-based access control (RBAC) system supporting 3 permission tiers, reducing unauthorized access incidents and improving admin configuration efficiency.
- Built a payment-splitting and coupon management module using Nest.js and PostgreSQL.
- Implemented JWT-based authentication and authorization, strengthening application security.
- Developed responsive frontend interfaces in React.js with Tailwind CSS, improving page load performance and UI consistency across devices.
- Optimized backend performance by introducing Redis-based queuing and containerized deployments, reducing average API response time by 10%.
- Increased frontend code reliability by writing component tests with Vitest.
- Refactored shared components into reusable, structured modules, cutting duplicate code by 20% and accelerating feature delivery.

Software Developer Intern — Qlith Innovation *Jul 2025 – Sep 2025*

- Built a Salon Management System from the ground up using React.js, Express.js, and MongoDB, enabling end users to manage bookings and staff schedules.
- Designed and implemented RESTful APIs and MongoDB schemas, supporting core booking and inventory workflows.
- Collaborated with a small development team to translate business requirements into functional frontend and backend features.
- Applied foundational full-stack principles (HTML, CSS, JavaScript, React) to deliver a production-ready web application under mentorship.

Software Developer Intern — Vault of Codes *Mar 2025 – Apr 2025*

- Contributed to collaborative development workflows using Git and GitHub, participating in pull requests.
- Integrated transactional email functionality using Nodemailer and Mailtrap, enabling automated notifications.
- Implemented Google API integrations (Google Sheets, Google Maps), automating data workflows and reducing manual data entry.

PROJECTS

Multi-Vendor E-Commerce Platform — Personal Project — *React.js, Nest.js, PostgreSQL, Redis, Docker, AWS EC2*

- Architected a multi-vendor e-commerce platform supporting user management, product variants, order management, and cart functionality.
- Integrated Razorpay payment gateway with webhook-based transaction verification.
- Implemented BullMQ job queues and Redis caching to offload asynchronous tasks, improving API throughput.
- Deployed the application via Docker containers on AWS EC2, establishing a repeatable, production-style deployment pipeline.

Salon Management System — Personal Project — *React.js, Express.js, MongoDB, JWT*

- Designed and built a full-featured salon management platform covering appointment booking, staff scheduling, service catalog, and customer records, replacing manual/paper-based scheduling for the target business.
- Implemented a role-based access control system with JWT authentication and authorization, securing access across 2 distinct user roles (admin and staff) and protecting sensitive customer and business data.
- Built a real-time appointment booking flow with conflict detection to prevent double-booking of staff and time slots, improving scheduling reliability.
- Designed normalized MongoDB schemas for bookings, staff, and services to support efficient queries and scale cleanly as the service catalog grew.
- Built a responsive, mobile-friendly React.js interface, enabling salon staff to manage bookings from any device with minimal training.

Retrieval-Augmented Generation (RAG) Application — Personal Project — *Next.js, LangChain, QdrantDB*

- Built a RAG pipeline using LangChain and QdrantDB, implementing vector embeddings, text splitting, and similarity search to power context-aware AI responses.
- Optimized vector database queries for semantic search across documents.

EDUCATION

B.Tech, Computer Science Engineering — Gandhi Institute for Education and Technology, Odisha CGPA: 8.11 | Expected 2026

Higher Secondary (CHSE, XII) — Bidwan Residential College 78.5% | 2022

Secondary (HSE, X) — HRM High School 66.6% | 2020